

Systematic Review and Meta-Analysis

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Abbreviations:

DAI, dietary antioxidant index; FRAP, ferric reducing antioxidant power; GDM, gestational diabetes mellitus; ORAC, oxygen radical absorption capacity; PCOS, polycystic ovary syndrome; PE, pre-eclampsia; TAC, total antioxidant capacity; TEAC, trolox equivalent antioxidant capacity; TRAP, total radical capture antioxidant parameter; VCE, vitamin C equivalent antioxidant capacity.

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Diet's total antioxidant capacity and women's health: systematic review and meta-analysis

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Abstract

The primary causes of female mortality often involve diseases related to oxidative stress. Dietary total antioxidant capacity (TAC) evaluates its antioxidant content and potential health effects. This study, registered with PROSPERO (ID: CRD42024427784), explores the association between dietary TAC and women's health outcomes, including endocrine conditions with gynaecological implications, obstetric outcomes, gynaecological conditions and oncological diseases related to the female reproductive system. We conducted a systematic search in MEDLINE (via PubMed), EMBASE, LILACS and CINAHL for observational studies published up to February 2024 that explored the relationship between dietary TAC and these health conditions. Data were analysed using RevMan 5.4 software. Nineteen studies met the eligibility criteria (sample sizes: 64–3209 women) and examined various conditions, including neoplasms (breast, endometrial and ovarian), bacterial vaginosis, menopause, polycystic ovary syndrome (PCOS), pre-eclampsia (PE), gestational diabetes mellitus (GDM), miscarriage, infertility and inflammation and oxidative stress markers. The meta-analysis identified a significant association between dietary TAC, measured in vitamin C equivalents, and breast cancer, revealing that women with the disease had a lower dietary TAC due to reduced antioxidant intake. Mixed results were found for endometrial cancer, while higher TAC levels were associated with a lower risk of PCOS and infertility. Among postmenopausal women, higher TAC correlated with fewer symptoms such as sleep issues and anxiety. In gestational conditions, higher dietary TAC was linked to a lower risk of miscarriage, GDM and PE. Twelve of the nineteen studies demonstrated significant associations between dietary TAC and the outcomes of interest.

Women account for approximately 50 % of the world's population; however, in light of the expanded concept of health, social gender discrimination towards them and the overload of responsibilities worsen this population's well-being and make it more susceptible to illnesses, fact that increases women's morbidity and mortality rates^(1–3). Accordingly, breast neoplasms and complications related to pregnancy and childbirth are among the main causes of mortality in the female population⁽⁴⁾.

These and other diseases' pathophysiology has been related to oxidative stress, which is a mechanism observed when the body presents an imbalance between oxidising and antioxidant compounds. It is worth noticing that the antioxidant defense system comprises endogenous enzymes, including superoxide dismutase, catalase and glutathione peroxidase, as well as exogenous compounds, such as vitamin C and lycopene, that can be ingested through food intake⁽⁵⁾. Furthermore, researchers have dedicated themselves to assessing the positive effect of diets rich in antioxidants on diseases' prevention and progression, as well as in controlling signs and symptoms of these pathologies. Researchers also observed that there is a high correlation between a diet's total antioxidant capacity (TAC) and plasma TAC in young adults has been already observed^(6,7).

A diet's antioxidant status can be assessed through dietary TAC, which is a dietary pattern analysis method. It is adopted to point out a diet's antioxidant content and its cumulative and synergistic effects since it can show populations' total antioxidant status^(8,9). The concept of dietary TAC was introduced as a new tool to assess dietary antioxidants' effects since it has proven to be as effective in measuring diet quality as other established diet quality indices because it can report inverse associations between dietary TAC and oxidative stress⁽⁹⁾.

Thus, if one takes into consideration the high prevalence of disorders affecting the female population every year, the association of oxidative stress and dietary TAC with these disorders' improvement or prevention, and the scarcity of systematic reviews seeking to investigate it, the aim of the present systematic review and meta-analysis was to assess the association between

dietary TAC and clinical conditions affecting women's health, including endocrine conditions with gynaecological implications, obstetric outcomes, gynaecological conditions and oncological diseases related to the female reproductive system.

Methods

Protocol and registration

This review was elaborated based on the document proposed by the Meta-analysis of Observational Studies in Epidemiology⁽¹⁰⁾ reporting checklist. The search for scientific articles, as well as data extraction and methodological quality assessment, was carried out by two independent investigators – disagreements between appraisers were resolved through consensus. Research formal registration was carried out at the PROSPERO database (ID: CRD42024427784).

Search strategy

The search for original observational studies assessing the association between dietary TAC and women's health outcomes was conducted to identify articles according to previously established inclusion criteria. The search was carried out in the following databases: MEDLINE (via PubMed), EMBASE, LILACS and CINAHL and through the reference list of articles obtained, until 29 February 2024. The following search MESHES were adopted: 'dietary total antioxidant capacity', 'woman' and 'women's health'.

These meshes were selected based on the previous readings of published articles relating diet TAC to health outcomes, as well as on MESHES selected for each assessed clinical condition. It is important to notice that all studies selected through this search were included in the present review. Publication year filters were not used.

Study eligibility criteria

This systematic review applied PICOS (Population, Intervention, Comparison, Outcome and Study Design) criteria, as detailed in Table 1. The following inclusion criteria were adopted: (a) studies assessing diet TAC in pregnant women, in women at reproductive age and women at pre- and post-menopausal periods, be them healthy or carrying diseases related to women's health – including endocrine conditions with gynaecological implications (e.g. polycystic ovary syndrome (PCOS)), obstetric outcomes (e.g. gestational diabetes mellitus (GDM), preeclampsia (PE), miscarriage), gynaecological conditions (e.g. infertility, menopause) and oncological diseases related to the female reproductive system (e.g. breast, ovarian and endometrial cancers) –; (b) studies conducted on human beings and (c) manuscripts published in English, Portuguese and Spanish about the herein assessed topic. Studies focused on diseases that are not specific to endocrine, obstetric, gynaecological or oncological outcomes, such as cardiovascular, bone, ocular and psychological diseases, as well as review articles, clinical trials, animal studies, and studies carried out *in vitro* were excluded from the review.

Study selection and data extraction

Articles selection was carried out through title and abstract analysis and full-text reading by two authors of the current study separately. Duplicates were excluded after combining findings from the adopted bases; subsequently, studies were selected by titles. The

Table 1. PICOS criteria for inclusion of studies

Parameter	Inclusion criteria
Population	Pregnant women, women at reproductive age, and women with pre- and post-menopausal periods
Exposure	Diet evaluated by the dietary total antioxidant capacity, regardless of the method used for analysis
Comparison	NA
Outcome	Gynecological, obstetric, endocrine, and oncological diseases or complications, such as breast/endometrial cancer, polycystic ovary syndrome (PCOS), pre-eclampsia, miscarriage, and infertility.
Study design	Observational studies

Abbreviations: NA, not applicable.

selected articles were subjected to abstract reading, and the potentially relevant ones were read in full. The corresponding author was contacted to request access to the full article when it was not available to go on with the assessments.

The following information was extracted from the selected articles: author; publication year and country where the research was carried out; study objective and design; sample size (*n*); features of women included in the research (whether they were healthy pregnant women, adult women carrying a disease, among others); participants' mean age; adopted dietary TAC assessment method; dietary TAC mean and standard deviation and main analysed outcomes.

Methodological quality assessment

Methodological quality assessment of the included studies was carried out in JBI's critical appraisal tool⁽¹¹⁾, which, in turn, is widely recommended for assessing observational studies. It is also worth highlighting that this tool analyzes three main domains and their sub-classifications by taking into consideration aspects linked to sample selection, comparability and determination of exposure/outcome of interest, besides its easy application and interpretation.

Statistical analyses

Meta-analyses were performed with studies that were carried out with similar pathologies and that presented similar methodology features in the evaluation and measuring units of dietary TAC to standardise the collected data. Means and standard deviations were calculated for each investigated dietary TAC analysis methodology by using the inverse variance method and the DerSimonian and Laird random effects model. Data heterogeneity was analysed through the Cochran Q test and I^2 statistic. All analyses were carried out in RevMan 5.4 software (The Nordic Cochrane Centre, The Cochrane Collaboration, Denmark).

Results

Search

Search in the databases led to 9269 studies and 2082 of them were duplicates. In total, 7156 of the remaining 7187 studies were excluded from the review at the title-reading stage. The reviewers' analysis included twenty-two studies in the full-text assessment. Two (2) texts were excluded from the study after the reading because they were conference abstracts, and one text was excluded

Table 2. Features of studies assessing diet's total antioxidant capacity and women's health outcomes, including endocrine, gynaecological, obstetric and oncological conditions, 2024

Author (publication year), country	Study Objective	Study design	Sample size	Age	Women's features	Assessment method	Used dietary survey	Total dietary antioxidant capacity value	Total dietary anti-oxidant capacity sd	Main outcomes
Giffkins et al., (2012), USA	Investigating the association between endometrial cancer risk and individual antioxidant micronutrients from food types and supplements, as well as the association between endometrial cancer risk and dietary TAC	Case-control	417 cases and 395 controls	Not described	Women with endometrial cancer (cases) and healthy women (controls)	ORAC (umolTE/100 g), TP (mgGAE/100 g) and FRAP (umolTE/100 g)	FFQ	H-ORAC: 13 058 (cases) e 13 809 (controls); L-ORAC: 429 (cases) and 442 (controls); T-ORAC: 13 358 (cases) and 14 133 (controls); TP: 1805 (cases) and 1895 (controls); FRAP: 5428 (cases) and 5669 (controls)	H-ORAC: 341 (cases) e 351 (controls); L-ORAC: 15 (cases) and 15 (controls); T-ORAC: ± 348 (cases) and 358 (controls); TP: 47 (cases) and 48 (controls); FRAP: 146 (cases) and 150 (controls)	Decreased disease risks were found for the highest TP intake tertile in comparison to the lowest one (OR: 0.62; CI 95 %: 0.39, 0.98. <i>P</i> = 0.02)
Giffkins et al., (2012), USA	Assessing the association between dietary TAC/ and individual intake of antioxidants (vitamin C, vitamin E, beta-carotene, selenium, lutein and lycopene) and the risk of ovarian cancer	Case-control	205 cases and 390 controls	Not described	Women with ovarian cancer (cases) and healthy women (controls)	ORAC (umolTE/100 g), TP (mgGAE/100 g) and FRAP (umolTE/100 g)	FFQ	H-ORAC: 13 000 (cases) and 13 488 (controls); L- ORAC: 394 (cases) and 437 (controls); T-ORAC: 13 275 (cases) and 13 809 (control); TP: 1776 (cases) and 1821 (controls); FRAP: 5361 (cases) and 5446 (controls)	H-ORAC: 504 (cases) and 361 (controls); L- ORAC: 20 (cases) and 14 (controls); T-ORAC: 514 (cases) and 368 (control); TP: 71 (cases) and 50 (controls); FRAP: 214 (cases) and 153 (controls)	There was little evidence of association between dietary TAC and risk of ovarian cancer (H- ORAC <i>P</i> = 0.65; L-ORAC <i>P</i> = 0.48; T-ORAC <i>P</i> = 0.96; TP <i>P</i> = 0.54; FRAP <i>P</i> = 0.66)
Pantavos et al., (2014), Netherlands	Assessing whether dietary TAC and individual dietary antioxidant intake were associated with breast cancer risk	Prospective cohort	3209	68 ± 39 years	Initially healthy women: 199 developed breast cancer	FRAP (mmol/d)	FFQ	15.18 (low tertile), 19.81 (medium tertile) and 24.48 (high tertile)	Not described	Individuals with moderate and high dietary FRAP at baseline (≥ 18.01 mmol/d) had a lower risk of developing breast cancer in comparison to individuals with low dietary FRAP (≤ 18.00 mmol/d) after adjustments made in potential confounders (OR (high v. low): 0.68; 95 % CI: 0.49, 0.96 and OR (medium v. low): 0.71; 95 % CI: 0.51, 0.99)

Table 2. (Continued)

Karimi et al., (2015), Iran	Investigating the association between dietary TAC and the risk of breast cancer	Case-control	100 cases and 175 controls	Cases: 46.2 ± 8.9 years; Controls: 45.9 ± 9.4 years	Women with breast cancer (cases) and healthy women (controls)	ORAC (μmolTE/100 g)	FFQ	1564 (cases) and 1584 (controls)	452 (cases) and (controls)	Consumption of fruits with TAC > 2030 (μmolTE/100 g) (OR: 0.16; 95 % CI: 0.06, 0.44, <i>P</i> < 0.01) and vegetables with TAC > 617 (μmolTE/100 g) (OR: 0.32; 95 % CI: 0.15, 0.71 in the third quartile and OR: 0.29; 95 % CI: 0.13, 0.68 in the fourth quartile, <i>P</i> < 0.01) had significantly reduced risk of breast cancer
Rossi et al., (2016), Italy	Investigating the role of dietary TAC in endometrial cancer risk	Case-control	454 cases and 908 controls	Cases: 60 (range 18–79 years; Controls: 61 (range 19–80 years)	Women with incident endometrial cancer (cases) and acute non-neoplastic conditions (controls)	FRAP (mmol/d), TEAC (mmol/d) and TRAP (mmol/d)	FFQ	FRAP: 22.72; TEAC: 7.37; TRAP: 10.41	FRAP: 10.49; TEAC: 3.16; TRAP: 5.24	Dietary TAC was inversely related to risk of endometrial cancer ((OR: 0.69, 95 % CI 0.47, 1.00, <i>P</i> : 0.04 for FRAP), (OR: 0.68, 95 % CI 0.46, 0.99, <i>P</i> = 0.02 for TEAC) and (OR: 0.68, 95 % CI 0.47, 0.98, <i>P</i> = 0.03 for TRAP) (quartile 1 v. quartile 4))
Sheikhi et al., (2017), Iran	Assessing the association between dietary TAC, serum TAC, and PE risk in women with PE and normal pregnancy	Case-control	55 cases and 93 controls	Cases: 29.3 ± 5.5 years; Controls: 26.6 ± 6.0 years	Pregnant women with PE (cases) and healthy pregnancies (controls)	ORAC (μmolTE/100 g) and TP (mgGAE/100 g)	FFQ	H-ORAC: 14 383.1 (cases) and 17 405.7 (controls); L-ORAC: 151.0 (cases) and 166.5 (controls); T-ORAC: 14 817.9 (cases) and 20 284.5 (controls); TP: 1173.1 (cases) and 1585.3 (controls)	H-ORAC: 795.0 (cases) and 1204.8 (controls); L-ORAC: 9.3 (cases) and 14.1 (controls); T-ORAC: 1236.1 (cases) and 1873.3 (controls); TP: 78.1 (cases) and 118.3 (controls)	After adjustments made in the main risk factors, no relationship was found between the H-ORAC (OR: 0.51; 95 % CI: 0.25, 1.05), L-ORAC (OR: 0.64; 95 % CI: 0.30, 1.33), TP (OR: 0.76; 95 % CI: 0.36, 1.59), and T-ORAC (OR: 0.04; 95 % CI: 0.23, 0.99) intake and risk of PE.
Abshirini et al., (2018), Iran	Assessing the association between dietary TAC and	Cross-sectional	188 cases and 212 controls	Cases: 55.5 ± 10 years; Controls:	Postmenopausal women without (controls) and	ORAC (μmolTE/100 g)	FFQ	24 438.9 (cases) and 25 757.2 (controls)	6727.9 (cases) and 7159.3 (controls)	There was reduction in hot flashes and sweating (OR: 0.51;

(Continued)

Table 2. (Continued)

Author (publication year), country	Study Objective	Study design	Sample size	Age	Women's features	Assessment method	Used dietary survey	Total dietary antioxidant capacity value	Total dietary anti-oxidant capacity sd	Main outcomes
	menopausal symptoms in middle-aged postmenopausal women			57 ± 7.7 years	with (cases) acute symptoms					95 % CI: 0.23, 1.13; $P = 0.03$), sleep issues (OR: 0.29; 95 % CI: 0.12, 0.68, $P = 0.003$), anxiety (OR: 0.38; 95 % CI: 0.16, 0.90, $P = 0.02$), exhaustion and difficulty concentrating (OR: 0.21; 95 % CI: 0.06, 0.69, $P < 0.004$). However, a positive association was observed between dietary TAC and sexual issues: OR rose from 1.72 to 2.47 and $P = 0.03$
Daneshzad et al., (2020), Iran	Investigating the association between dietary TAC and GDM	Case-control	200 cases and 263 controls	Cases: 29.27 ± 5.96 years; Controls: 27.61 ± 6.34 years	Pregnant women with GDM (cases) and healthy pregnancies (controls)	TRAP (mmol/d), FRAP (mmol/d) and TEAC (mmol/d)	24-hour recall	TRAP: 8.34 (cases) and 8.60 (controls); FRAP: 11.13 (cases) and 12.88 (controls); TEAC: 7.18 (cases) and 7.45 (controls)	TRAP: 0.26 (cases) and 0.23 (controls); FRAP: 0.28 (cases) and 0.25 (controls); TEAC: 0.20 (cases) and 0.17 (controls)	GDM risk was lower among those in the highest FRAP tertile (OR: 0.15; 95 % CI: 0.08, 0.29, $P < 0.0001$). There was no significant association between GDM risk and TRAP (OR: 1.62; 95 % CI: 0.94, 2.79, $P = 0.072$) and TEAC (OR: 1.56; 95 % CI: 0.89, 2.72, $P = 0.113$)
Safabakhsh et al., (2020), Iran	Assessing the association between dietary TAC and incidence of breast cancer among Iranian women	Case-control	150 cases and 150 controls	24–73 years (age-matched cases and controls)	Women with newly diagnosed breast cancer (cases) and healthy (controls)	ORAC (μmol/d)	FFQ	25 906.8 (cases) and 27 074.3 (controls)	9471.0 (cases) and 10 711.1 (controls)	No association was observed between dietary TAC and the incidence of cancer among women before and after adjustments ($P = 0.28$ and $P = 0.91$, respectively), nor when they were subdivided into pre ($P = 0.27$, after adjustments

Table 2. (Continued)

										<i>P</i> = 0.62) and post-menopause (<i>P</i> = 0.78, after adjustments <i>P</i> = 0.77). The association between dietary TAC and breast cancer was OR: 1.00 (0.20, 2.98) in tertile 2 and OR: 0.77 (0.23, 4.26) in tertile 3 (<i>P</i> = 0.91)
Vahid et al., (2021), Iran	Assessing the association of NQI and DAI with the chances of miscarriage	Retrospective case-control	67 cases and 68 controls	Cases 30.72 ± 4.80 years; Controls 31.06 ± 4.24 years	Women with a history of ≥ 3 miscarriages before the 20th week of pregnancy (cases) who were still pregnant at 20 weeks, and who did not have a miscarriage (controls)	DAI	FFQ	Values were stratified into DAI < 0.054 (indicating low antioxidant intake) and ≥ 0.054 (indicating high antioxidant intake)	Not described	DAI protective effect against miscarriage was found in comparison to cases and controls ((OR = 0.82; 95 % CI: 0.70, 0.97. <i>P</i> = 0.021), and DAI as a categorical variable, thus indicating low or high antioxidants intake (OR DAI < 0.054 v. DAI ≥ 0.054 = 0.43; 95 % CI: 0.20, 0.91; <i>P</i> = 0.027
Sasanfar et al., (2021), Iran	Determining the association between dietary TAC and breast cancer risk among Iranian women	Case-control	412 cases and 456 controls	Cases: 46.3 ± 10.4 years; Controls: 44.2 ± 11.3 years	Women with breast cancer (cases) and healthy women (controls)	FRAP (mmol/100 g)	FFQ	11.3 (cases) and 12.1 (controls)	5.8 (cases) and 7.9 (controls)	Individuals in the highest dietary TAC quartile were less likely to have breast cancer than those in the lowest quartile (OR: 0.61; 95 % CI: 0.38, 0.99. <i>P</i> = 0.06). Postmenopausal women with higher dietary TAC had lower breast cancer odds than those with lower dietary TAC (OR: 0.47; 95 % CI: 0.24, 0.93, <i>P</i> < 0.05). This association strengthened after additional adjustments made in BMI (OR: 0.28;

(Continued)

Table 2. (Continued)

Author (publication year), country	Study Objective	Study design	Sample size	Age	Women's features	Assessment method	Used dietary survey	Total dietary antioxidant capacity value	Total dietary anti-oxidant capacity sd	Main outcomes
										95 % CI: 0.12, 0.72, $P < 0.05$)
Noormohammadi et al., (2022), Iran	Investigating the association between adherence to the Mediterranean diet and dietary TAC with BV	Case-control	143 cases and 151 controls	Cases: 30 (25–33) years; Controls: 32 (24–37) years	Women affected by BV (cases) and healthy women (controls)	FRAP	FFQ	Quartile 1 = 1.00; Quartile 2 = 0.70; Quartile 3 = 0.99	Quartile 2 = 0.38–1.28; Quartile 3 = 0.55–1.78	There was no significant association between dietary TAC and BV
Shoaibinobarian et al., (2022), Iran	Assessing dietary TAC and its association with chances of PCOS in Iran women	Case-control	303 cases and 588 controls	Cases: 29.1 ± 5.9 years; Controls: 28.8 ± 6.2 years	Women with (cases) and without (controls) a history of PCOS	ORAC (mmol/d)	FFQ	11.9 (cases) and 14.1 (controls)	3.9 (cases) and 4.3 (controls)	The highest tertile of dietary TAC was associated with reduced likelihood of PCOS (OR: 0.81; 95 % CI: 0.59, 0.96, $P = 0.038$)
Han et al., (2022), South Korea	Testing the hypothesis that dietary TAC is inversely associated with cancer recurrence and mortality in patients with breast cancer	Prospective cohort	66 with the disease and 537 without disease recurrence	With the disease: 49.7 ± 10.3 years; Without the disease: 52.2 ± 10.5 years	Patients with breast cancer who underwent surgery 5 years before the research	VCE (mgVCE/d)	FFQ	355.6 (cases) and 476.1 (controls)	237.1 (cases) and 363.1 (controls)	Dietary TAC was inversely associated with the risk of cancer recurrence before ($P = 0.008$) and after ($P = 0.042$) adjustments made in confounding factors. However, it was not related to mortality ($P = 0.141$ before and $P = 0.432$ after adjustments)
Maugeri et al., (2022), Italy	Assessing the association of DAI and IBD with the prevalence of high-grade CIN	Cross-sectional	539 women	40.2 ± 10 years	Women with abnormal Papanicolaou test, who underwent high-risk human papillomavirus (HPV) screening and histological test through colposcopy	DAI	FFQ	Not described	Not described	A lower rate of HPV-positive women was observed among those with higher DAI ($P < 0.001$); the index was not associated with CIN2 diagnosis or with more serious injuries (OR (95 % CI): 2nd tertile 1.18 (0.70, 2.01), $P = 0.532$; 1st

Table 2. (Continued)

Jalali et al., (2023), Iran	Investigating the association between dietary TAC and the risk of breast cancer	Case-control	136 cases and 272 controls	Cases: 49.5 ± 10.7 years; Controls: 47.1 ± 10.1 years	Women with newly diagnosed breast cancer (cases) and hospitalised (controls)	ORAC (mmol/d) and FRAP (mmol/d)	FFQ	ORAC: 15.2 (cases) and 16.1 (controls); FRAP: 13.3 (cases) and 13.9 (controls)	ORAC: 6.1 (cases) and 7.0 (controls); FRAP: 8.6 (cases) and 7.3 (controls)	tertile 1-12 (0.62, 2.00), $P = 0.711$) Dietary TAC was not related to breast cancer odds ((ORAC: OR Q4-Q1 = 1.01; 95 % CI = 0.42, 2.44, $P = 0.96$) and (FRAP: OR Q4-Q1 = 1.04; 95 % CI = 0.53, 2.05, $P = 0.8$))
Vahid et al., (2023), Iran	Investigating the DAI/ chances of breast cancer/ correlation between DAI and pathobiological markers association	Case-control	145 cases and 148 controls	Tertile 1: 49.7 ± 12.7 years, tertile 2: 48.7 ± 11.2 years; tertile 3: 49.1 ± 11.8 years	Women with incidence of breast cancer (cases) and hospitalised (controls)	DAI	FFQ	Values were expressed as < -0.6 and ≥ -0.6 for sample stratification	Not described	A significant association between cancer odds and DAI (OR = 0.18; 95 % CI: 0.09, 0.37; $P < 0.01$). Modeling DAI as a continuous variable with multivariable control showed insignificant results (OR = 1.00; 95 % CI: 0.98, 1.02; P -value = 0.59)
Kabodmeri et al., (2023), Iran	Assessing the link between female infertility and DAI	Case-control	125 cases and 125 controls	Cases: 37.12 ± 50 years; Controls: 32.92 ± 41 years	Women diagnosed with reduced ovarian reserves (cases) and with normal ovarian reserve (controls)	DAI	FFQ	-0.57 (cases) and 0.65 (controls)	3.49 (cases) and 5.34 (controls)	DAI was negatively associated with infertility (OR: 0.94, 95 % CI: 0.88, 0.97, $P = 0.03$). The association remained significant after making adjustments in age, BMI, underlying diseases, fertility frequency, IVF failure and caloric intake.
Case-control										

(Continued)

Table 2. (Continued)

Author (publication year), country	Study Objective	Study design	Sample size	Age	Women's features	Assessment method	Used dietary survey	Total dietary antioxidant capacity value	Total dietary anti-oxidant capacity sd	Main outcomes
Alkan et al., (2023), Türkiye	Assessing dietary TAC, DII, and inflammation and oxidative stress serum biomarkers in women with breast cancer		32 cases and 32 controls	Cases: 45.4 ± 9.5 years; Controls: 45.1 ± 8.5 years	Women diagnosed with breast cancer for the first time (cases) and healthy women (controls)	ORAC (umolTE/100 g), FRAP (umolTE/100 g), TEAC (mmol/TE), TRAP (mmol/TE) and VCE (mg VCE/d)	3-day food record	T-ORAC: T1 = 13 765.0, T2 = 16 315.5, T3 = 15 120.8, T4 = 15 133.5 and control = 17 323.8; L-ORAC: T1 = 919.2, T2 = 1280.9, T3 = 1202.1, T4 = 1344.9 and control = 1232.8; H-ORAC: T1 = 12 928.6, T2 = 16 315.5, T3 = 13 651.9, T4 = 14 157.8 and control = 161 171.2; FRAP: T1 = 5.225, T2 = 5.95, T3 = 5.125, T4 = 5.275 and control = 8.175; TEAC: T1 = 4.0, T2 = 4.2, T3 = 4.1, T4 = 3.6 and control: 5.9; TRAP: T1 = 4.4, T2 = 4.7, T3 = 4.1, T4 = 4.4 and control = 8.3; VCE: T1 = 529.8, T2 = 608.6, T3 = 520.2, T4 = 544.9 and control = 565.1	T-ORAC: T1 = 4134.0, T2 = 6578.4, T3 = 5148.8, T4 = 7075.9 and control = 6966.8; L-ORAC: T1 = 721.2, T2 = 721.9, T3 = 581.2, T4 = 903.1 and control = 593.7; H-ORAC: T1 = 3361.3, T2 = 6578.4, T3 = 4060.3, T4 = 6615.6 and control = 7528.2; TEAC: T1 = 2.2, T2 = 2.5, T3 = 3.1, T4 = 2.8 and control = 4.3; TRAP: T1 = 3.4, T2 = 2.5, T3 = 2.8, T4 = 3.6 and control = 4.9; VCE: T1 = 338.9, T2 = 271.9, T3 = 312.8, T4 = 309.1 and control = 387.3	There was no significant association between dietary TAC values and serum TAS, TOS and OSI values at Tercile 1, Tercile 3, and Tercile 4. A weak positive correlation was found between dietary TAC (VCE) and serum TAS level at Tercile 2 (rho = 0.371, P = 0.036)

Abbreviations: AFD, Antioxidant Food Database; BV, Bacterial vaginosis; CIN, cervical intraepithelial neoplasia; DAI, Dietary antioxidant index; DII, Dietary Inflammatory Index; FRAP, ferric reducing antioxidant potential; GDM, gestational diabetes mellitus; H-ORAC, hydrophilic oxygen radical absorption capacity; HPV, human papillomavirus; L-ORAC, lipophilic oxygen radical absorption capacity; NQI, nutritional quality index; ORAC, oxygen radical absorption capacity; PE, pre-eclampsia; TAC, total antioxidant capacity; TAS, total antioxidant status; TE, Trolox equivalents; TEAC, antioxidant capacity equivalent to Trolox; T-ORAC, total oxygen radical absorption capacity; TOS, total oxidant status; TP, total phenolics; TRAP, total radical scavenging antioxidant parameter; T1, before surgery; T2, before chemotherapy; T3, in the 6th month of chemotherapy; T4, in the 12th month of chemotherapy; VCE, vitamin C equivalents.

because its full-text access was not allowed, not even after request to the corresponding author. In total, nineteen studies met the eligibility criteria defined for this systematic review and they remained in the study. Five (5) of them had their data included in the meta-analysis (Figure 1).

Features of analysed studies

Fifteen of the nineteen selected articles followed the case-control design^(8,12–25), two of them were cohort studies^(26,27) and two were cross-sectional research^(9,28). The sample size ranged from 64 to 3·209 women. Twelve (12) studies were carried out with Iranian participants^(8,9,14–17,19,20,22–25), two with Italian women^(18,28), two studies were carried out with American women^(12,13), one research comprised Korean women⁽²⁶⁾, one article regarded residents in the Netherlands⁽²⁷⁾ and one comprised women from Turkey⁽²¹⁾. The main characteristics of the included studies are shown in Table 2.

Eleven (11) studies assessed dietary TAC among women with neoplasms, seven studies assessed women with breast cancer^(14,16,19,20,25–27), three research assessed women with endometrial cancer^(12,18,28), and one article regarded women with ovarian cancer⁽¹³⁾. One study analysed women with bacterial vaginosis⁽¹⁷⁾, one research assessed women with menopausal participants⁽⁹⁾ and one research involved women with PCOS⁽²²⁾. Two studies analysed pregnant women (one was about women with PE⁽²³⁾ and the other about mothers with GDM⁽⁸⁾), one study assessed women with a history of miscarriage⁽²⁴⁾ and another analysed female infertility⁽¹⁵⁾. One study related diet TAC to inflammation and oxidative stress serum markers in women with breast cancer⁽²¹⁾. The studies used clinical examinations and diagnoses to identify the assessed comorbidities.

Five studies used oxygen radical absorption capacity (ORAC) assays to determine dietary TAC^(9,16,19,22,23), four studies used the dietary antioxidant index (DAI) to do so^(15,24,25,28), two studies used the total radical capture antioxidant parameter (TRAP) in this analysis, as well as ferric reducing antioxidant power (FRAP) and Trolox equivalent antioxidant capacity^(8,18), three studies only used FRAP assays^(17,20,27), three studies analysed the combination of FRAP and ORAC methods^(12–14), one study adopted the vitamin C equivalent antioxidant capacity (VCE) method⁽²⁶⁾ and one article analysed all the aforementioned methods, except for DAI⁽²¹⁾. In addition, twelve of the nineteen included studies showed a close association between dietary TAC and the outcomes of interest^(9,12,15,16,18,20,22,24–28).

When analysing the main outcomes of the included studies, it is possible to note that, for the most part, dietary TAC showed an influence on the prevention of the most diverse diseases related to women's health in general. In this context, four of the seven studies assessing breast neoplasia found that women with this cancer type recorded reduced dietary antioxidants intake in comparison with those without this disease. There was a significant inverse correlation between dietary TAC and breast cancer after adjustments made in the main confounding factors, yet, a significantly lower consumption of TAC was also found among women who had a recurrence of the disease^(25,26).

Furthermore, three studies showed that the inverse association between dietary TAC and breast cancer risk became non-significant, after controlling the potential confounding factors, even if women are divided into pre- and post-menopausal states^(14,16,19). Around endometrial neoplasia, only one of the three analysed studies observed an association between antioxidant

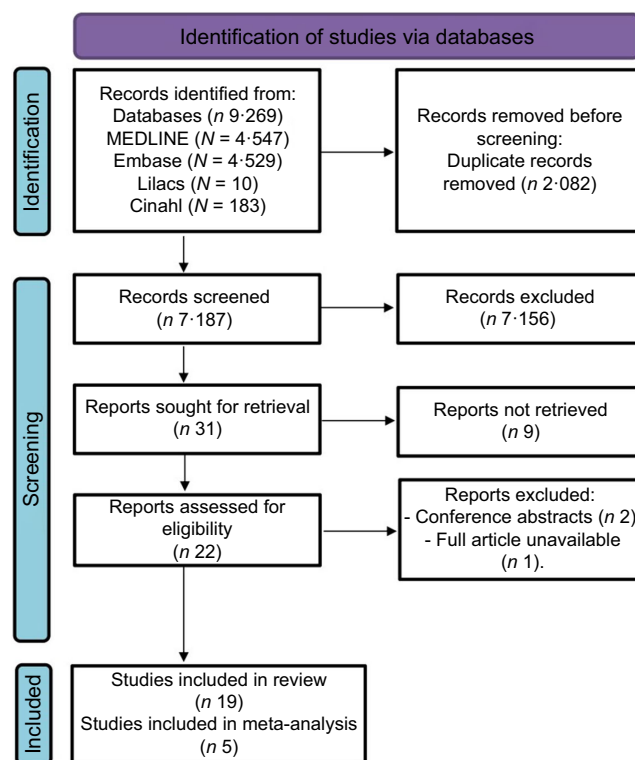


Figure 1. Flow chart of search and selection process.

index and endometrial lesions⁽²⁸⁾, and one author did not find an association through the FRAP method, despite there being a reduction in the risk of cancer in the group with higher intake of total phenolic compounds⁽¹²⁾.

On endometrial cancer, Rossi et al.⁽¹⁸⁾ observed that diet TAC is inversely related to the risk of the disease. In turn, Gifkins et al.⁽¹³⁾ found little evidence of the association between dietary TAC and ovarian neoplasia development. Alkan et al.⁽²¹⁾ only observed a weak positive association between dietary and serum TACs in women with breast cancer, before chemotherapy.

Studying PCOS, Shoaibinobarian et al.⁽²²⁾ concluded that there is an association between those recording higher dietary TAC and with lower chance of having PCOS in comparison to female participants with and without PCOS diagnosis. Regarding infertility, Kabodmehri et al.⁽¹⁵⁾ also showed that women with higher DAI had a lower risk of infertility, with an inversely significant association. Around postmenopausal women, the highest TAC level were less likely to experience symptoms, such as sleep issues, anxiety, hot flashes and sweating, whereas those accounting for the lowest level were more likely to experience sexual issues⁽⁹⁾. Furthermore, a study evaluated the influence of dietary TAC and the occurrence of bacterial vaginosis; however, no relationship was observed between them⁽¹⁷⁾.

Regarding diseases of the gestational period, the study of the relationship between the occurrence of abortion and dietary TAC observed a significant protective effect in women who have shown the highest level of dietary TAC⁽²⁴⁾. Furthermore, one study looked at pregnant women and assessed those with and without GDM, where they observed that the occurrence of the disease was greater in women with lower antioxidant intake, just as a similar finding was found in cases of "PE"^(8,23).

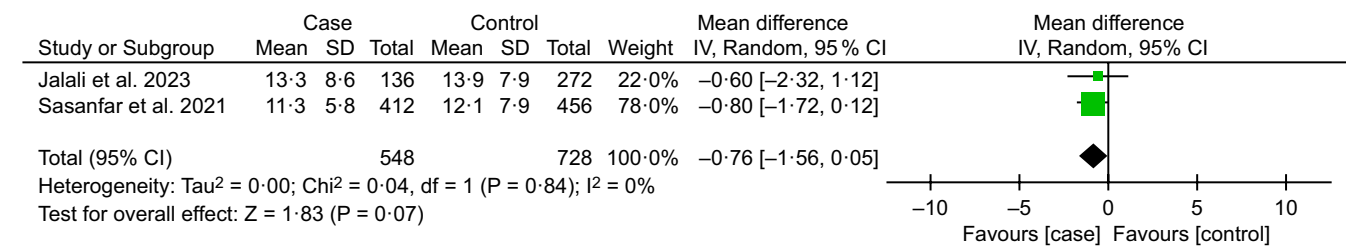


Figure 2. Forest plot of studies using the FRAP method to assess diets' total antioxidant capacity, expressed in mmol/d. FRAP, ferric reducing antioxidant power.

Studies' methodological quality

About assessing the methodological quality of articles analysed in JBI's critical appraisal tool, all studies about neoplasms, infections, PCOS, GDM, PE, infertility and serum biomarkers met the tool's requirements, so their quality was considered adequate. On the other hand, one study²⁴ did not meet important tool requirements; therefore, it was considered a low-quality article. It did not make the exemption of the exposure of interest in the control group clear, and it would have directly interfered with the recorded results, besides not reflecting the results' complete reliability.

Meta-Analysis

FRAP: Two studies were included in the meta-analysis: 548 women with breast cancer (case group) and 728 healthy women as the control group. FRAP was not significantly different between studied groups (mean difference: -0.76, CI 95 % (-1.56, -0.05), *P* = 0.07; I² = 0 % and *P* = 0.84) (Figure 2).

ORAC: Two studies were included in the meta-analysis: 182 women with breast cancer in the case group and 182 healthy women in the control group. No significantly different ORAC value was observed in the groups (standard mean difference: -0.31, CI 95 % (-0.79, 0.17), *P* = 0.20; I² = 68 % and *P* = 0.08) (Figure 3).

VCE: Two studies were included in this analysis: 98 women with breast cancer in the case group and 569 healthy women in the control group. Vitamin C equivalents' levels were significantly lower in the case group than in the group of individuals without diseases (mean difference: -110.53, CI 95 % (-171.53, -49.52), *P* < 0.001; I² = 0 % and *P* = 0.38) (Figure 4).

Discussion

This systematic review, complemented by meta-analysis, provides a comprehensive view of the association between dietary TAC and various health conditions affecting women. Although not all studies were included in the quantitative analysis due to methodological reasons, the findings demonstrate that dietary TAC levels are significantly lower in women with diseases compared to the control group. This outcome reinforces dietary antioxidant intake's influence on preventing conditions affecting women's health. In addition, diversity in study designs highlights the broad approach adopted to explore this association, as well as sample size extension – ranging from 64 to 3209 women – which raises questions about the general profile and robustness of the current findings. These results make it crucial to take into consideration the representativeness of the assessed population.

Studies' geographic distribution shows the majority of research was carried out with Iranian participants, as well as with Italian, American and Korean women and with residents in the

Netherlands and Turkey, which raises questions about cultural and environmental factors' influence on the diet TAC/female health association. Accordingly, the analysis applied to the assessed populations highlights the global exploration of the association between dietary TAC and several women's health conditions, such as the history of miscarriage, bacterial vaginosis, menopause, PCOS, complicated pregnancies (PE and GDM), infertility and cancer (breast, endometrium and ovary).

Four of the seven studies assessing breast neoplasia found that women with this cancer type recorded reduced dietary antioxidants intake in comparison to those without this disease. There was a significant inversed correlation between dietary TAC and breast cancer after adjustments made in the main confounding factors, yet, a significantly lower consumption of TAC was also found among women who had a recurrence of the disease^(25,26). Furthermore, the number of participants in the highest dietary TAC quartile was 0.39 times less likely to have the aforementioned disease than those in the lowest quartile⁽²⁰⁾. Pantavos et al.⁽²⁷⁾, in their turn, noticed that moderate and high dietary FRAP (≥18.01 mmol/d) led to a lower risk of developing breast cancer than low dietary FRAP after adjustments made to potential confounding factors.

However, three studies showed that the inversed association between dietary TAC and breast cancer risk became non-significant, after controlling the potential confounding factors, even if women are divided into pre- and post-menopausal states^(14,16,19). Only one of the three analysed studies on endometrial neoplasia observed an association between antioxidant index and endometrial lesions⁽²⁸⁾. Another author has not found this association through the FRAP method, but cancer risk was reduced among the highest tertile of total phenolic compound intake in comparison to the lowest one, after using the ORAC database and making adjustments in the main covariates⁽¹²⁾.

Rossi et al.⁽¹⁸⁾ also found correlations, such as diet TAC inversely related to the risk of endometrial cancer and risk between the lowest and highest quartile, in all analysed methods (FRAP, Trolox equivalent antioxidant capacity and TRAP). However, Gifkins et al.⁽¹³⁾ found little evidence of an association between dietary TAC and ovarian neoplasia development. Alkan et al.⁽²¹⁾ only observed a weak positive association between dietary and serum TACs in women with breast cancer, before chemotherapy, when it comes to the association among dietary TAC and inflammation serum biomarkers (tumour necrosis factor alpha, interleukin 1β and interleukin 6) and oxidative stress (malondialdehyde, protein carbonyl and total oxidising state).

Shoaibinobarian et al.⁽²²⁾ concluded that there is an association between those recording higher dietary TAC and with lower chance of having PCOS in comparison to female participants with and without PCOS diagnosis. Kabodmehri et al.⁽¹⁵⁾ also showed that women with higher DAI had a lower risk of infertility, with an

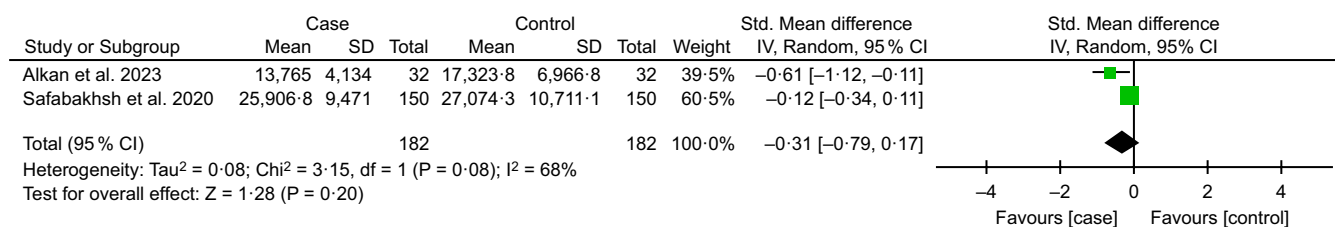


Figure 3. Forest plot of studies using ORAC method to assess diets' total antioxidant capacity expressed in $\mu\text{molTE}/100\text{ g}$. ORAC, oxygen radical absorption capacity.

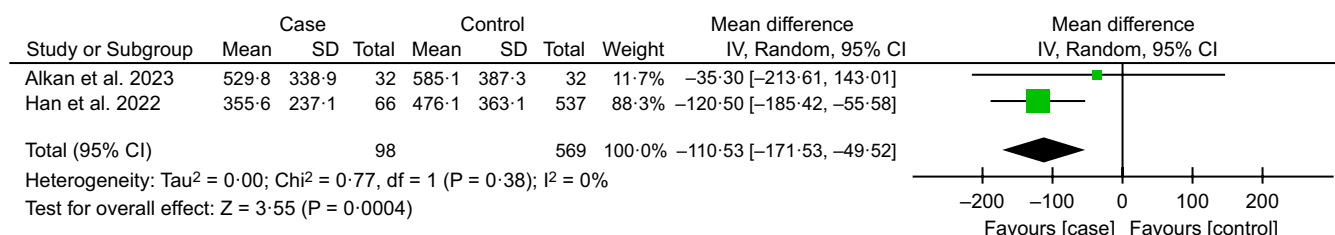


Figure 4. Forest plot of studies using the Vitamin C Equivalent method to assess diet's total antioxidant capacity.

inversely significant association. Likewise, a cross-sectional study carried out with postmenopausal women assessed the dietary TAC/menopausal symptoms association and observed that women presenting the highest TAC level were less likely to experience symptoms, such as sleep issues, anxiety, hot flashes and sweating, whereas those accounting for the lowest level of it were more likely to experience sexual issues⁽⁹⁾.

Researchers have shown a positive association between menopausal symptoms and increased risk of chronic diseases and oxidative stress, mainly due to oestrogen deprivation. This outcome highlights the importance of controlling these symptoms through likely modifiable factors, such as diet and overall intake of antioxidant nutrients^(9,29,30). The possible explanation for such mechanisms linking antioxidant intake to menopausal symptoms reduction lies in higher antioxidant status caused by diets rich in these nutrients, which leads to increased dietary TAC. Furthermore, these diet types count on abundant amounts of phytoestrogens lignans and isoflavones, which are plant-derived oestrogens less powerful than estradiol, but that are capable of binding to the oestrogen receptor and reducing hot-flashes severity and frequency^(9,31,32).

PCOS is primarily an endocrine disorder with gynaecological implications, where oxidative stress plays a key role its development, due to insulin resistance and hyperglycaemia commonly associated with this condition⁽²²⁾. Therefore, researchers have pinpointed that oxidative stress decrease is positively related to more mature oocytes in infertile PCOS women. Thus, food types rich in antioxidants can protect the body from oxidative damage in these patients. Recent data also suggest that high dietary TAC can reduce PCOS changes^(22,33,34).

A case-control study aimed at assessing the DAI/chances of miscarriage association and observed significant protective effects in women who showed the highest level of dietary TAC in comparison to those who did not, after crude and multivariate data adjustments were made⁽²⁴⁾. Regarding PE, women with the disease had significantly lower dietary TAC than pregnant women without this comorbidity, after adjustments made to the main risk factors, such as energy and pre-pregnancy BMI. Nonetheless, no association was observed between TAC intake and PE risk⁽²³⁾.

The literature points out that women with GDM present increased oxidative stress biomarkers and reduced antioxidant activity after the analysis of the association between gestational disorders, such as PE and GDM, and dietary TAC. This finding suggests an imbalance during pregnancy, as shown in the study by Mandani et al. in 2021⁽³⁵⁾. Two studies looked at pregnant women and assessed those with and without GDM. They found that FRAP intake was significantly lower in the case group than in the control and that GDM risk was 85 % lower among women in the highest FRAP tertile. However, there was no significant association between GDM risk and TRAP and Trolox equivalent antioxidant capacity methods⁽⁸⁾.

Another case-control study sought to investigate the dietary TAC/bacterial vaginosis association. However, vaginosis odds were not associated with total antioxidant intake. This finding disregarded age, BMI, waist circumference, number of cigarettes per day, pregnancy frequency and exercise⁽¹⁷⁾.

These results raise important reflections to help better understand dietary TAC and its implications for women's health outcomes, to provide a solid starting point for more in-depth discussions about this complex association and its clinical outcomes. Furthermore, the whole variety of assessed conditions highlights the complexity of dietary TAC's influence on women's health, and it suggests that different antioxidant mechanisms may play specific roles in distinct contexts.

Five articles were suitable for the quantitative meta-analysis because they presented similar, or likely standardisation methodologies and pathologies. The included studies produced current evidence of the association between dietary TAC and breast cancer. Breast cancer demonstrated strong epidemiological evidence when dietary TAC was assessed using vitamin C equivalents, showing that the total antioxidant intake in women with the disease is significantly lower than in healthy women. This suggests that a low consumption of antioxidant-rich foods may be associated with a higher risk of developing this oncological condition. Conversely, when dietary TAC was evaluated using the FRAP and ORAC methods, no significant associations were observed.

The findings highlight a critical relationship between dietary TAC and breast cancer risk, underscoring the potential role of

antioxidants in disease prevention. These dietary TAC/neoplasms association is well established in the literature. The meta-analysis carried out with twenty-one studies, and another one conducted with nineteen studies, have concluded that high dietary TAC has a protective effect against these diseases, since it reduces the incidence of breast, endometrial, colon, gastric and lung cancer^(36,37). The pathophysiology of neoplasms can explain this association, since it involves the secretion of cytokines and inflammatory mediators, such as TNF- α and leptin. This process increases oxidative stress and tissue damage⁽³⁸⁾; therefore, antioxidant vitamins act in ruling out/neutralising reactive oxygen species, increasing antioxidant capacity and reducing cellular oxidative damage⁽³⁹⁾.

The observed lower dietary TAC in women with breast cancer could reflect dietary patterns characterised by insufficient consumption of fruits, vegetables and other antioxidant-rich foods. This underscores the importance of dietary quality in cancer prevention strategies and highlights a potential area for public health interventions aimed at promoting antioxidant-rich diets. Moreover, the results may also suggest that vitamin C equivalents capture relevant aspects of antioxidant intake more effectively than other methods, such as FRAP and ORAC, again raising questions about the variability of these assays in assessing dietary antioxidants, as they focus on distinct chemical reactions and target compounds. These discrepancies also could highlight the complexity of the antioxidant system, where interactions between multiple compounds and pathways influence the overall protective effect.

Future research should delve deeper into the methodological differences between these assessments to determine which most accurately reflects biologically active antioxidant capacity related to cancer risk. In addition, longitudinal studies and intervention trials are needed to explore whether increased consumption of antioxidant-rich foods reduces the incidence or progression of other diseases related to women's health, as few studies have been conducted on this topic, limiting scientific knowledge on the subject.

These findings also emphasise the need for personalised dietary recommendations, as individual variations in metabolism, genetics and lifestyle can modify the impact of antioxidants on cancer risk. Understanding these nuances may pave the way for more targeted approaches to prevention, ensuring that dietary strategies are effective and tailored to individual needs.

It is crucial to recognise and consider some limitations inherent to the assessed studies, such as the diversity of methods used to determine dietary TAC, including assays like ORAC, DAI, TRAP, FRAP, Trolox equivalent antioxidant capacity and VCE, after analysing the results in the present systematic review, complemented by meta-analysis. This finding underscores the need for methodological standardisation, as the chosen methods can influence the consistency of results and comparability between studies.

Conclusion

It is possible to conclude that, despite methodological differences used in studies included in the present literature review, dietary TAC was lower in individuals in the disease groups than in those belonging to the control group. This finding reflects the consequences of eating habits based on low fruit and vegetable consumption, which are the main antioxidant-source food types and it also highlights the relevance of ingesting these compounds in regular diets to prevent conditions affecting health, both in the short- and long terms.

Furthermore, it is recommended that further studies focus on a deeper understanding of the mechanisms underlying the effects of dietary TAC on women's health conditions and that health policies be developed to promote adequate consumption of foods that are sources of antioxidants, aiming to improve health and prevent diseases.

Supplementary material. For supplementary material/s referred to in this article, please visit <https://doi.org/10.1017/S0007114525000443>

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The authors declare no conflict of interest.

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